HumanFactor — Code Summary for the Team

What the R / Jupyter notebook actually does, and what it actually found — BMAT202P Probability & Statistics Lab, Milestone 1 (~30–40% of the full project)

Purpose of this document

This is a plain-language walkthrough of the executed notebook `HumanFactor_Analysis.ipynb`, written for teammates who need to understand what the code does without re-reading all 83 cells. Every number quoted below was produced by running the notebook against the real dataset — nothing here is invented.

1. Dataset

File: `student_lifestyle_dataset.csv` **Source:** Kaggle (secondary data — we did not collect this ourselves)

- 2000 student records, 8 columns.
- Columns: Student_ID, Study_Hours_Per_Day, Extracurricular_Hours_Per_Day, Sleep_Hours_Per_Day, Social_Hours_Per_Day, Physical_Activity_Hours_Per_Day, GPA, Stress_Level.
- 0 missing values, 0 duplicate rows, 0 impossible values (verified directly in code, not assumed).
- Notable finding: the five “hours per day” columns always sum to almost exactly 24.0 for every student. This means the dataset was likely constructed/derived rather than purely independently measured — it's a real finding from the code, documented honestly in the notebook's Limitations section rather than hidden.

2. Environment & project layout

The analysis runs in **R** (not Python) inside a Jupyter notebook using the IRkernel. A dedicated conda environment (`humanfactor`) was created with R 4.5.3 and the packages: tidyverse, janitor, psych, corrplot, GGally, gridExtra, knitr, kableExtra, scales.

Folder structure:

- `data/raw/` — original Kaggle file, never modified.
- `data/processed/` — type-corrected and cleaned CSVs produced by the code.
- `notebooks/HumanFactor_Analysis.ipynb` — the executed notebook (this document summarizes it).
- `outputs/figures/` — 11 saved plots. `outputs/tables/` — 7 saved result tables.
- `report/` and `presentation/` — written outlines for the final thesis and demo deck.

3. What the code actually does, step by step

Step 1 — Import & inspect (Experiment 1)

Reads the CSV with `readr::read_csv()`, prints dimensions, column names, R data types, first/last rows. Converts `Stress_Level` into an ordered factor (Low < Moderate < High) since it's genuinely ordinal, not just text. Exports a type-corrected copy to `data/processed/humanfactor_typed.csv`.

Step 2 — Data quality checks & cleaning

Checks for missing values, duplicate rows/IDs, negative or out-of-range values, inconsistent category labels, and IQR-based outliers — all programmatically, not by eyeballing. Result: the file needed almost no cleaning (0 missing, 0 duplicates). Standardizes column names with `janitor::clean_names()` and saves `data/processed/humanfactor_cleaned.csv`.

Step 3 — Summary statistics & tabulation (Experiment 2)

Computes N, mean, median, variance, SD, min, Q1, Q3, max, IQR for every numeric column, and a frequency/percentage table for `Stress_Level`. Saved to `outputs/tables/descriptive_statistics.csv` and `frequency_tables.csv`.

Step 4 — Visualization

Generates 9 individual plots (histograms of sleep/GPA/study/physical-activity hours, a bar chart of stress levels, 3 scatterplots of lifestyle variables vs GPA with a fitted trend line, and a boxplot of GPA by stress category), plus a correlation heatmap and a final regression plot — 11 PNGs total, all saved under `outputs/figures/`.

Step 5 — Correlation analysis

Computes the Pearson correlation of every lifestyle variable against GPA and ranks them by strength, so the strongest predictor is chosen by the code, not hand-picked in advance.

Step 6 — Simple linear regression + R^2 (Experiment 3)

Fits `GPA ~ strongest_predictor` using R's `lm()`, reports intercept, slope, and R^2, plots the regression line, and checks residuals (residuals-vs-fitted, residual histogram) as a basic sanity check on the model's assumptions.

Step 7 — Preliminary multiple regression (Experiment 4, Phase 1 only)

Fits `GPA ~ all 5 lifestyle-hour variables` together, reports coefficients, Multiple R^2, and Adjusted R^2, and flags that the predictors are expected to be somewhat intercorrelated because of the 24-hour constraint found in Step 2. This step is explicitly labelled **preliminary / not final** in the notebook.

Step 8 — Coverage matrix for Experiments 5–11

Rather than implementing the remaining 7 experiments, the code builds a table stating, for each one, whether this dataset can support it and why (or why not), and what would be needed if not (e.g. a real auxiliary dataset for the paired t-test, Poisson fit, RBD, and Latin Square — none of which this dataset's structure can legitimately support).

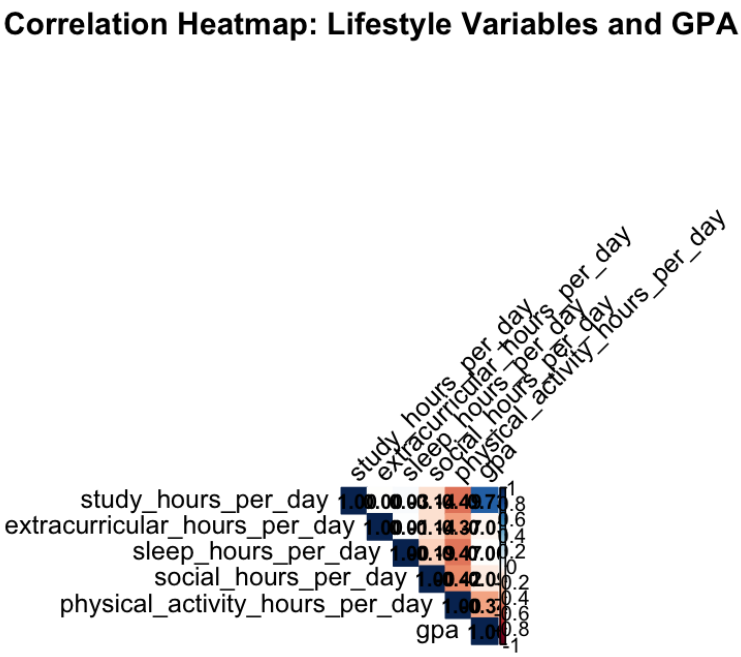
4. Key results produced by the code

These are the actual numbers the notebook calculated — not estimates.

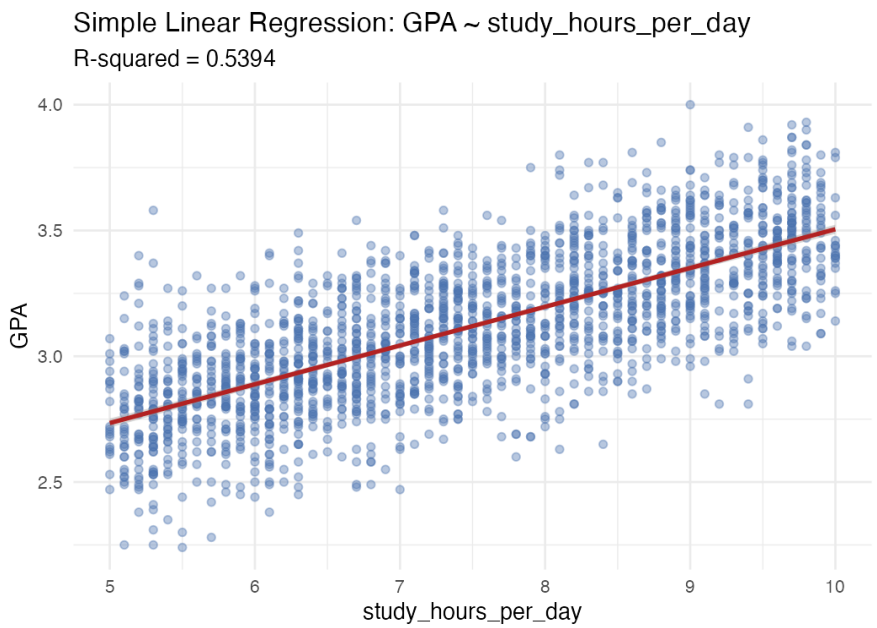
Metric	Value	What it means
Strongest predictor of GPA	Study_Hours_Per_Day	Chosen automatically as the variable with the largest |correlation| with GPA.
Correlation (r) with GPA	0.734	A strong positive linear association.
Simple regression R^2 (GPA ~ Study Hours)	0.539	~54% of the variation in GPA is linearly explained by study hours alone.
Multiple regression Multiple R^2	0.541	Adding all 5 lifestyle variables together only raises explained variance to ~54.1%.
Multiple regression Adjusted R^2	0.540	Confirms the extra 4 predictors add almost nothing once study hours is already included.

Reading this honestly: *study time is clearly the dominant lifestyle factor linked to GPA in this dataset. Sleep, extracurricular, social, and physical-activity hours add almost no extra explanatory power once study hours is accounted for — likely because they're all competing for the same fixed 24-hour budget. None of this implies that studying more causes a higher GPA — the data is observational, not experimental.*

5. Example output figures



Correlation heatmap — every lifestyle variable's linear relationship with GPA and with each other.



Simple linear regression of GPA on Study_Hours_Per_Day, with R^2 shown in the plot subtitle.

6. BMAT202P syllabus coverage (honest status)

The full lab has 11 required experiments. This milestone finishes 3 of them, starts a 4th, and documents a concrete plan for the rest — nothing is pretended to be done.

#	Experiment	Status
1	Data types, import/export	COMPLETED
2	Summary statistics, tabulation, visualization	COMPLETED
3	Correlation + simple linear regression + R^2	COMPLETED
4	Multiple linear regression	PARTIAL (Phase 1 model only)
5	Binomial distribution fit	PLANNED — no binary variable exists yet
6	Normal + Poisson distribution	PLANNED — Poisson needs an auxiliary count dataset
7	One-sample mean / proportion test	PLANNED
8	Two-sample means / proportion test	PLANNED
9	Independent + paired t-test	PLANNED — paired test needs auxiliary data
10	Chi-square GOF + contingency	PLANNED
11	ANOVA: CRD / RBD / Latin Square	PLANNED — RBD & Latin Square need auxiliary data

7. Limitations the code itself surfaced

- Secondary, observational data — no causal claims anywhere in the analysis.
- The exact 24-hour sum across lifestyle variables means they are not statistically independent predictors.
- No missing data at all is unusual for genuine self-report data, suggesting this Kaggle file may be partly simulated for practice rather than a raw real-world survey — disclosed rather than glossed over.
- No binary, count, or paired/blocked-design variables exist, so Experiments 5, 6 (Poisson), 9 (paired), and 11 (RBD/Latin Square) will need a separate, properly sourced auxiliary dataset.

8. What's next

- Finish Experiment 4 properly (multicollinearity/VIF check, model refinement).
- Run the one-way ANOVA on GPA by stress level (Experiment 11's CRD component) using this same dataset.
- Source a legitimate auxiliary dataset for the Poisson fit, paired t-test, RBD, and Latin Square designs.
- Write the final Abstract, Conclusion, and References sections of the Mini Thesis once all 11 experiments are done.

Full code: `notebooks/HumanFactor_Analysis.ipynb`. *All tables referenced above are in* `outputs/tables/`.